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On full-grade *-*ro*- formations in Greek and Indo-European

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1. Introductory.

1.1. Primary "*-*ró*- adjectives" in PIE normally exhibit zero grade of the root and accent on the suffix – a commonplace observation hardly requiring elaborate demonstration or reference. This pattern, symbolized here as **R(Ø)-ró-** (R = root), can be exemplified by very many forms, including such familiar items as **h₁rud*^{h}*-ró-* 'red' (ἐρυθρός, Lat. *ruber* etc.) or **duh₂-ró-* 'far, long' (Ved. *dūrá-*, Lat. **dūro-* [→ denom. *dūrāre* 'extend']). Beside the latter, however, appears Hom. δηρός 'long' (together with its cognate Arm. *erkar* 'id.'), apparently reflecting a full-grade **dṷeh₂-ro-*; a question arises, then, as to the status of such **R(E)-ro-** forms (E = *e*-grade or *o*-grade; accent left unspecified at this point), especially given the equally commonplace assumption that thematic formations involving suffixes like *-*ro*- "have no ablaut" (Beekes 1995:190).[1]

1.2. The handbooks[2] do little more than acknowledge the existence of **R(E)-ro-** forms;[3] more often (though still only occasionally), such forms are invoked to support analyses leading to or requiring the assumption of similar formations. Among recent examples, note J. Katz' interpretation of OLat. BERBER in the *Carmen Arvale* in terms of a meaning like 'steadfast', via a full-grade **d*^{h}*erĝh-ro-* – cf. the corresponding "normal" zero grade in Av. *dərəzra-* 'firm' < **d*^{h}*rĝh-ro-*

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[1] I use δηρός here partly for expository convenience (cf. Janda, in §1.2. below, and further discussion of δηρός in §8.), under the assumption that IE **duh₂-ró-* will not yield Pr.-Gk. **dṷāro-* directly; but those who operate with a broad version of the "Lex Francis-Normier" (according to which a sequence **duh₂-ró-* will indeed yield **dṷāro-*; for discussion see e.g. Klein 1988, Rasmussen 1990/91, Clackson 1994:41ff.) can for the moment substitute any number of other **R(E)-ro-** forms (numerous examples below) not affected by this phonological indeterminacy. The specific profile of parallel **R(Ø)-ró-/R(E)-ro-** *adjectives*, however, is rare; more on this point below.

[2] A particularly useful treatment in Wackernagel-Debrunner II.849-59, cf. also the main section of Frisk 1934 ("Das indogermanische *-ro-*Suffix im Altindischen", pp. 1-49).

[3] E.g. Wackernagel-Debrunner II.853, with minimal descriptive commentary of the sort "Guṇa in *ghorá-* und kl. *smera-* ... Auch in den verwandten Sprachen erscheint vereinzelt Hochstufe".

(1998:216n96). Similarly, M. Janda (1996) seeks to justify an etymology of Gk. *νεβρός* 'fawn' in terms of a full-grade **negʷ-ro-* "'defenseless, *naked*', based on the word for 'naked' seen in Ved. *nagná-* and other forms (including another full-grade *ro*-form in Arm. *merk*), the morphology being supported by the observation "[f]ür die Hochstufe im *ro*-Adjektiv vgl. nhd. *Eiter* < **h₂eid-ro-* ..., δηρός < **dṷeh₂-ro-*" (1996:91-2). On an einzelsprachlich level, X. Tremblay observes (1996:113, contra Pirart 1992:21ff.) that Av. *aš.vandra-* need not be emended (as Pirart had suggested), since "le degré plein se rencontre en védique aux racines mediae *-n-*: *á-tandra-* 'infatigable', *candrá-* 'lumineux', etc."[1]

1.3. In a more theoretical vein, R. Wachter (1997:10f.) cites several **R(E)-ro-** forms within the context of the chronological development of accent and ablaut patterns from "early" to "late" PIE, whereby archaic forms with "single full grade" like **h₁rud^h-ró-* are systematically opposed to forms with more than one full grade of the type **h₁leu̯d^h-ero-* 'free' (Venet. *louderobos* etc.), which Wachter takes to reflect, generally speaking, a later stratum of derivation (even though **h₁leu̯d^h-ero-* and similar forms are reconstructable for some stage of the proto-language). Without denying the theoretical interest of viewing PIE accent/ablaut theory within such "früh-, mittel-, spätgrundsprachlich" frameworks (cf. Dunkel et al. 1994), speculations of this sort do not bring us much closer to understanding the morphological status, within reconstructable patterns of PIE suffixal derivation, of a **R(E)-ro-** form like **dṷeh₂-ro-*.

2. Notes on data to be excluded (and included).

2.1. In view of the relative neglect of the pattern under consideration, the first step is to adduce a sufficient quantity of relevant data for analysis. It is important, however, to be aware of certain difficulties associated with some of the data types involved, which leads us to exclude from consideration, at least provisionally, a number of forms and form types.

Perhaps the most basic problem concerns the potential ambiguity between **-ro-* and **-r-o-*: thus, as Wachter has correctly pointed out (1997:12f.), it is strictly speaking not possible to make this distinction even for forms like **udro-* 'watery'. Despite the standard assumption that such a form must be derivationally based on an *r/n*-stem (and thus reflect **-r-o-* historically), we cannot actually exclude an **ud-ro-*, formed with one or another **-ro-* suffix. For heuristic purposes, then, it seems best to exclude **R(E)-ro-** forms that may be based on *r*-stems or *r/n*-stems.

2.2. Insofar as we are concerned, in the first instance, with primary formations, we exclude for the moment formations that are clearly secondary, such as **temH-s-ro-* (Lat. *tenebrae* 'darkness' etc.), **h₂erĝ-u-ro-* (*ἄργυρος* 'silver', with "Caland system" associations), **h₁os-u-ro-* (*οὖρος* 'fair wind', Nussbaum 1998a:152), and many others. Nevertheless, formations based on root nouns are worth considering, since it is not always easy to distinguish secondary derivatives of root nouns from primary formations – cf. δηρός itself, beside δήν 'for a long time', Hitt. *duwān* 'far', these being traditionally interpreted to reflect an acc. sg. **dṷéh₂-m* of a root noun (somewhat differently Dunkel 1997:70f., in terms of a "root-instrumental" in **-m*).[5]

2.3. The following categories of forms should also be excluded, for fairly evident reasons:

2.3.1. R(E)-ru- and **R(E)-ri-** forms lacking a corresponding **-ro-* (even if – indeed, partly because – the derivational pattern **-ro-* → **-ru-/*-ri-* itself is fairly clear; especially for **-ro-* → **-ri-*, see Weiss 1996:204f., Nussbaum 1999, Melchert 1999:366f.);

2.3.2. R(ĕ)-ro- forms to "Narten" (acrostatic) roots, e.g. **h₁éd-ro-* (the hypothetical basis [cf. §2.3.1. !] of Hitt. *ēdri-* n. 'food'), or **(h₁)ĕĝʷʰ-ro-* (cf. Lat. *ēbrius* 'drunk');

2.3.3. R(E)-ero- forms (even if partly found beside corresponding forms with plain **-ro-*), as in some terms of relationship et sim.: **h₁leu̯d^h-ero-* (§1.3.), **su̯ekᵘ-ro-* 'father-in-law', **b^hend^h-(e)ro-* (*πενθερός* 'father-/son-in-law', Li. *beñdras* 'companion'), etc.

2.4. Beyond the above, literally scores of otherwise difficult or unusable forms have had to be excluded. To cite a convenient example, Ved. *śávīra-* 'mighty' (RV I.30.17b, I.32.2b, to which one might add several instances of trisyllabic "*śúū́ra-*", on which see recently Gippert 1997:77) looks descriptively like a **R(E)-ro-** form, and could thus reflect an old **ḱéu̯H-ro-* beside the **ḱuH-ro-* in *śū́ra-* 'heroic; hero'; indeed, this is precisely the assumption made by Gippert, who terms *śávīra-* a "full grade variant (**śáu̯āra-* < **ḱéu̯h₁ro-*)" vis-à-vis *śū́ra-*. But, apart from the somewhat suspicious attestation pattern of the rare *śávīra-/"śúū́ra-"* vs. *śū́ra-* (approx. 170x RV), S. Jamison's findings on laryngeal vocalization in Indic (1988) lead us to expect a **ḱéu̯H-ro-* to have developed rather as ***śávīra-*. Now, as Jamison has shown, the short *-i-* resulting from vocalized *-H-* not uncommonly ends up as *-ī-* secondarily, often as a result of metrical pressures. In the present case, *śávīra-* is in fact attested both times in the double-iambic form *śávīrayā*, of which one instance (I.30.17b) recovers the cadence of an octosyllable (cf. also Gippert on the cadential "*śúū́raḥ*"# at I.122.10d [11-syllable pāda]). Under these circumstances, it becomes at least equally probable that, as has sometimes been thought (see *EWAia* II.624, with references), *śávīra-* reflects a **śávira-* that is not

[1] A synchronic distributional fact noted by Wackernagel II.853, though without explanation; I return to this briefly below (§10.3.1.).

[5] Lindeman (1997:186) questions the root-noun interpretation of Hitt. *duwān*, but seems at least to accept that of δήν.

an old full grade at all, but merely a secondary creation to the superlative *śaviṣṭha-*, in which case *śavīra-*, despite superficial appearances, has no evidential value for the analysis of **R(E)-ro-** forms.

2.5. The merest sampling of a few similarly difficult or unusable forms might include items like the following, treated here in abbreviated fashion purely so as to illustrate certain recurring problems among such material.

2.5.1. The Myc. /leu̯kro-/ of PY Jn 415 (MN) *re-u-ko-ro-o-pu$_2$-ru* is traditionally taken as an error for /Leu̯ko-ophrūs/ (e.g. Lejeune 1972:143, *Docs.*[2]), even though "we should expect rather *Leukophrus*" (*Docs.*[2] s.v., cf. *re-u-ko-nu-ka* etc.). The form is apparently accepted as reflecting a genuine /leu̯kro-/ by Hajnal ("etymologisierende Schreibweise statt *<*re-u-ko-ro-pu$_2$-ru*>", 1995:137n173). At bottom, however, even if a Myc. /leu̯kro-/ 'white' actually existed, such a form could result from a secondary contamination of one sort or another (e.g. [Pr.-Gk.] **leu̯ko-* X **eruth*ro-, **k*^{h}*lōro-* → *leu̯kro-*, or **leu̯ko-* X **lukro-* [cf. Skt. *rucira-*] → *leu̯kro-*), and hence is of no real value.

2.5.2. For Gk. *ξηρός* 'dry', Lat. *serēnus* 'clear, dry'/*serēscō* 'become dry', OHG *ser(a)wēn* 'become dry' (see the discussion in Schrijver 1991:338, 353), interpretation in terms of a full-grade **kseh$_1$-ro-* accounts directly for *ξηρός*, but requires that the Latin and Germanic vocalism reflect a "Dybo's Law" shortening (thus explicitly Meiser 1998:75). While this is possible, the interest of this material is limited: the "root" **kseh$_1$-* is a hypothetical construct based solely on these forms, a situation which makes it extremely difficult to evaluate the nature of the *-ro-*formation.

2.5.3. Lat. *gibber* 'hunchbacked' may well show expressive gemination (see e.g. Leumann 1977:182); it is thus impossible to tell whether the form reflects an original zero grade (cf. *lippus* 'bleary-eyed', *crassus* 'thick', etc.) or (via the "*littera*-rule") a full grade, i.e. **gībe̯r* < **ge̯ib*^{h}*-ro-* (cf. Li. *geibùs* 'bulky, clumsy').

2.5.4. Gk. *κλῆρος* 'lot' (cf. OIr. *clár* 'plank'? see Kazansky and Falileev 1998) appears at first glance to be a zero-grade **kl̥h$_2$-ro-* to **kelh$_2$-* 'strike'. But this is problematic, insofar as a ***kl̥h$_2$-ro-* should have yielded * *κάλαρος*. If one could justify a "Schwebeablaut" variant **kleh$_2$-*, which seems to derive some support from *κλάω* (see Peters 1980:97), one could then argue for a **klḗh$_2$-ro-*, thereby yielding an isolated and potentially interesting **R(E)-ro-** form. But this picture is further complicated by the fact that *κλάω/κλαντ-* can simply be secondary to *κλάσσαι* (Schrijver 1991:174); on balance, then, *κλῆρος* is easier to interpret as a zero grade, though probably via a substantivized **kl̥h$_2$-ró-*, with Greek accent-retraction concomitant with the substantivization.

2.6. Conversely, certain types of forms should be included into our purview, even if they are not " **-ró-* adjectives" from a synchronic point of view, whether in individual languages or at the PIE level. Even beyond the simple fact that adjectives can be substantivized (in which case various **R(E)-ro-** substantives need to be examined), it is also well known that PIE contained more than one

type of **-ro-* formation beyond the " **-ró-* adjectives" (these largely with "Caland system" associations), most notably a series of substantival **-ro-* formations (generally lacking Caland system associations); and among these, at least some full grades are found, as in an apparent verbal noun like **déh$_3$-ro-* 'gift' (*δῶρον*, Arm. *towr*, OCS *darŭ*).[6]

2.7. The conclusion to be drawn from the above forms and discussion is that one must begin by examining a core of relatively clear-cut items, including certain non-adjectival and non-primary **R(E)-ro-** types. We begin with a case (or possibly two) involving secondary derivation from a root noun (cf. §2.2.), among other types of secondary derivation.

3. *mēms-ro-* (and *nas-ro-*?); *temH-s-ro-*.

3.1. Among the best cases of secondary **-ro-* based on root nouns (§2.2.) is the well-attested **mēms-ro-* (cf. **mēms-* 'flesh') of Lat. *membrum* 'limb', *μηρός* 'thigh-piece', OIr. *mír* 'portion', Slav. **mę̇zdro/*mę̇zdra* (RCS *mezdra/mjazdra* 'skin' etc.). The unmarked assumption would be that this simply reflects an adjectival **mēms-ro-* 'having to do with flesh, fleshy', which was then independently substantivized in Italic, Greek, Celtic and Slavic. Nevertheless, Hom. *μῆρα* 'thigh-pieces' has the look of an archaism (cf. also Slav. **mę̇zdra*), which at least authorizes the hypothesis (more economical in any case) that what may underlie these formations was in the first instance a collective **mḗms-reh$_2$* 'fleshy stuff, hunks of flesh' (> *μῆρα*, Slav. **mę̇zdra*, Lat. *membra*), with sg. *membrum*, *μηρός* etc. as back-formations.

3.2. In a similar way, one could suppose that the **nas-ro-* derived from **nas-* 'nose', as attested in Balto-Slavic (Ru. *nozdrjá* 'nostril' etc.[7]), reflects an original collective **nás-reh$_2$* 'nasal material' (> 'nasal passage(s)', cf. Li. pl. *nasraĩ* 'jaws, mouth', with a semantic shift). It remains, however, equally possible (if not, indeed, more attractive) to interpret this, with Nussbaum (1986:221f.), as a "delocatival" or "partitive/possessive" **-ro-* adjective ('at/on/in/of the nose').[8]

3.3. While not derived from a root noun, the secondary **temH-s-ro-* is likewise suggestive in showing what more properly seems to be a collective **témH-s-reh$_2$-* in addition to Lat. *tenebrae* (§2.2.), also Ved. *támisrā-*, Av. *tąθra-* (both also pl.) 'dark night, darkness' (cf. *EWAia* I.627). It is thus worth investigating the possibility that at least one of the functions of primary **-ro-* as it appears in full-grade formations involved original collectives.

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[6] On "substantival **-ro-*" as distinct from (Caland system) **-ro-* adjectives, see Nussbaum 1976:109f.

[7] On **-st-* > **-zd-* in both **mę̇zdra* and *nozdrjá*, see Ivanov 1998:328.

[8] To be sure, the more loosely defined "partitive/possessive" analysis could also apply to **mēms-ro-*, although a "delocatival" account is not attractive.

4. Some possible primary collectives.

4.1. The well-established and familiar **h₂eĝ-ro-* (Ved. *ájra-*, *ἀγρός*, Lat. *ager* 'field' etc.) embodies a number of difficult problems. On the semantic side, there is no general agreement as to exactly how the meaning of the term is to be conceived vis-à-vis **h₂eĝ-* 'drive' (see recently Anttila 1999 for extensive discussion of the semantics of the root and a number of its derivatives). Formally, the Vedic root accent does not match the Greek oxytonesis, and it has also been claimed (cf. §2.1.) that the form continues an original *r/n*-stem (thus e.g. Anttila, partly following Cowgill), although evidence for this is limited (see again Anttila 1999, passim, for discussion). In the present connection, at least as far as the Rig-Vedic material is concerned, one could imagine that the singular has been back-formed to an original collective meaning something like 'plains, countryside' (*ájra-* RV 9x, 8x pl., especially in what appears to be a traditional collocation in the form of a merism 'flatlands and mountains'), which at some early stage underwent an "animatization"; the accent could then be explained on the basis of an original root-accented collective. More directly, however, it seems particularly attractive to align Gk. *ἄγρα* 'the hunt; prey' with such a formation, where an original collective referring to 'that which is driven' makes good sense (see Anttila 1999:122ff. on *ἄγρα* in connection with **h₂eĝ-* in hunting usages, and further 130 on Celt. **agrom/*agrā* in OIr. *ár* 'carnage', W. *aer* 'battle').[9]

4.2. The full-grade **smei̯-ro-* traditionally reconstructed on the basis of material in Indic, Italic and Germanic (e.g. Ved. *ásmera-* 'not bashful'/Class. *smera-* 'smiling, friendly', Lat. *mīrus* 'wonderful', OE *gāl-smære* 'inclined to laughter, facetious', *smær(e)* 'lips') is also problematic in some respects, but could perhaps be considered in these general terms.

4.2.1. Lat. *mīrus*, to begin with, is not especially "adjectival": it is rarely attributive (soon replaced in this function by *mirabilis*), makes no superlative, and the comparative is all but unattested (see *DELL* s.v.). The oldest forms and usages involve a neut. *mīrum* (*mīrum est ...*, *mīrum quam ...*, *mīrum nī ...*, *nīmīrum*, etc.), and *mīror* 'be amazed' is generally taken as denominative to *mīrum/mīra*, not deadjectival to *mīrus* (see Flobert 1975:79n3). Thus *mīrus* is conceivably a secondary adjectivization of a neut. *mīrum/mīra*, based on a collective **smei̯-reh₂*. To be sure, the connection of *mīrus* with the Indic and Germanic material has sometimes been doubted, especially on semantic grounds;[10] but this would still leave the Germanic and Indic **R(E)-ro-** material.

[9] *ἄγρα* as collective vis-à-vis *ἀγρός*: also Rasmussen 1999:§4.4 (though with a somewhat different semantic interpretation). On the accentual relationship between the two cf. **§10.2.1.** below.

4.2.2. The Germanic forms are indeterminate in many ways. It seems clear, at any rate, that the *i*-umlaut of OE *smær(e)* (nom. pl. *smæras*, dat. *smærum*) points to something like a **smoi̯r(i)i̯o-* 'lip' as a secondary derivative (perhaps essentially agentive, i.e. 'smiler') to some prior *r*-formation; this, in turn, could have been a neut. **smói̯-ro-* 'laughter, smiling' (cf. Lat. *mīrum* above), with semantic traces perhaps appearing in some of the English material (in addition to OE *gāl-smære* cited above, note also *smæran* 'mock, deride' and ME *smēre* adv. 'happily, heartily', generally with *laughen*). I return below (§10.3.3.) to the question of the *o*-grade, which is at least to some extent a general Germanic feature.

4.2.3. RV *ásmera-* (II.35.4a) seems to mean something like 'ohne (verschämt) zu lächeln' (thus Geldner); in principle, this could derive from a negative bahuvrīhi 'lacking embarrassment, without giggling', based on a noun **smerā̆-*. In that case, the Class. Skt. adj. *smera-* (Hariv., Kālid. etc.) could simply have been extracted from an "adjectival" interpretation of RV *ásmera-*.

4.3. As for **déh₃-ro-* 'gift' (*δῶρον* etc., §2.6.): this could conceivably continue a collective **déh₃-reh₂* 'gift-age, gifts' (roughly > *δῶρα*), with back-formed neut. singular.[11]

4.4. For **h₂eḱ-ro-* (*ἄκρος* 'extreme, outermost, at the tip', OCS *ostrŭ* 'sharp'; indirectly Lat. *ācer* 'sharp', *acerbus* 'bitter', Ved. *áśri-* 'sharp edge, corner', etc.), a possible collective may appear directly (or almost directly) in *ἄκρα* 'tip, summit, citadel', beside which *ἄκρος* (note again the root accent) had already been suspected of being a secondary adjectivization, as persuasively argued by Frisk (1938:114).[12]

4.5. A full-grade **ĝʰeh₁-ro-* is reconstructable on the basis of Greek and Latin, note especially Hom. *χηρωσταί* 'those who plunder an inheritance' beside

[10] For an alternative proposal, see recently Weiss (1996:203n22), who suggests a **meis-o-* or **mis-ro-* 'shining' for Lat. *mīrus*, cf. Hitt. *mišriwant-* 'id.', ultimately (following a suggestion of G. Neumann) with Skt. *miṣáti* 'opens the eyes' (thus perhaps originally *'wide-eyed' or the like).

[11] One might, in this way, distinguish between a "true" verbal noun **déh₃-no-* (Lat. *dōnum* etc.) — an essentially substantival type of formation based on verb roots: **ǵégʰ-no-* 'conveyance', **sǵép-no-* 'sleep' etc. (although this point cannot be pursued further here) — and a parallel collective **déh₃-reh₂* (ultimately > *δῶρον* etc.), a situation which only secondarily produces what appear to be synonymous substantives **déh₃-no-*/"**déh₃-ro-*" 'gift' (as traditionally reconstructed).

[12] Somewhat similarly Ruijgh (1995:347n8): "Sous l'influence de *ἄκρα* 'cap' ..., interprété comme féminin singulier ..., **ἀκρός* 'pointu' a été remplacé par *ἄκρος*." But see also **§7.3.2.** below.

Lat. *hērēs* 'heir', originally 'eater(s) of that which is left behind' (Dunkel 1987; further Hom.+ *χήρη* 'widow' and other forms, all belonging with the root of Ved. *jáhāti* 'leaves behind', see *LIV* 152f.). Hom. *χηρωσταί* has already been taken to imply a neut. **χῆρον* 'abandoned property' (see e.g. *GEW* s.v. *χηρωσταί*); one might better assume rather a collective **χῆρα* (adjectival *χῆρος* 'bereaved, bereft' appears only relatively late [Eur.+]).

4.6. The *i*-stem of Lat. *febris* 'fever' is often thought to be a secondary substitution for an original *o*-stem (Leumann 1977:347, cf. *cīvis*, *fortis* etc.), in which case *febris* would point to a **dʰegʷʰ-ro-* (to **dʰegʷʰ-* 'burn'); but even if the *i*-stem is old within Latin, it would continue a (concretized) abstract **dʰegʷʰ-ri-*, which itself would have been based on an adjectival **dʰegʷʰ-ro-* in any case (cf. Nussbaum 1999). The antiquity of the formation is confirmed by Gk. *τέφρα* f. 'ashes', which again points in the direction of a collective.[13]

5. Some isolated forms.

5.1. It is not difficult to find relatively isolated (and thus perhaps archaic) collective-looking material reminiscent of that in §4.; thus, beside Gk. *ἄγρα*, *ἄκρα*, *τέφρα*:

5.1.1. (ἡ) *ἕδρα*/*ἕδραι* [Hom.+] 'seat, abode, session etc.' (partly pl. tantum);

5.1.2. (ἡ) *λέπρα* 'leprosy' [Hdt.+] (*'scabs, peelings', cf. *λέπω* 'peel'), adjectivized as MN *Λεπρός* 'Scabby' [Hippon.], otherwise adj. *λεπρός* Hipp., Theophr.+ only.

5.2. Full-grade **-ro-* formations are also found in Armenian (cf. Džaukjan 1982:131f.), some of which could be of this general type:

5.2.1. Arm. *sowr* 'sword, knife' (also adj. 'sharp') could continue a **ḱéh₃-reh₂* *'sharp(ened) material';

5.2.2. Arm. *sēr* 'love' (cf. **ḱe̯i̯-ɥo-* 'dear': Lat. *cīuis*, Ved. *śéva-* etc.) could continue a collective-derived abstract **ḱéi̯-reh₂*, although uncertainty about the precise meaning of the root hampers the analysis.

5.2.3. Arm. *lir* 'quantity, fullness' < **pleh₁-ri-* belongs with *πλήρης* 'full of, sated', *πληρόω* 'make full', traditionally assumed to be based on an *o*-stem **πληρο-*, cf. OLat. *plērus*/Class. *plērusque* 'very many, for the most part' and (OLat./Class.) adv. *plērumque* (OLat. also adv. *plērum*) 'for the most part'. The Greek vocalism, to be sure, is phonologically ambiguous between full grade and zero grade, and the Latin vocalism is, so to speak, "morphologically ambiguous", since *plē-* might here be a replacement for zero-grade *plā-* (cf. *plēnus* 'full', probably adjusted from the phonologically expected ***plānus* [< **pl̥h₁-nó-*] after *-plēre*, *-plētus*). But Arm. *lir* is difficult to derive from a zero grade; moreover, the "superlatival" and "substantival" meaning of the synchronic Greek and Latin adjectives is reminiscent of what was seen with *ἄκρα*/*ἄκρος* (**§4.4.**), a feature which an original collective of the sort **pléh₁-reh₂* (roughly) 'the full (amount)' (cf. adv. *plērum(que)*) might help explain.

5.3. Outside Greek and Armenian, similar material might include items like the following:

5.3.1. [**dém(h₂)-reh₂* ?] Gmc. **timra-* (OE/Eng. *timber* etc.) '(piece of) building material' (→ denominative verb, as in Go. *-timrjan*);

5.3.2. [**h₂éi̯d-reh₂* or **h₂éi̯d-reh₂* ?] Gmc. **aitra-* (OHG *eitar*/NHG *Eiter* etc.) 'pus' (*'swelling-material', cf. *οἰδέω* etc., *LIV* 230);

5.3.3. [**légʰ-reh₂* ?] OHG *legar* n. 'lying, resting-place, grave', Go. *ligrs* m. 'bed, couch' etc. (**légʰ-reh₂* 'lying area/material' [?], vs. the instrument noun **legʰ-tro-*: *λέκτρον*, OHG *lehtar*);

5.3.4. [**u̯ép-reh₂* ?] IIr. **u̯aprā-* (: Ved. *vápati* 'strews') in Skt. *vaprā-* [Sū.] 'fireplace', *vapra-* [Ep.+] m./n. 'earthen wall', YAv. *vafra-* m. 'snow' (Khot. *borā-*) etc. (*'strewage, strewn material');

5.3.5. [**ḱéh₂d-reh₂* ?] Av. *sādra-* n. 'woe, torment' (: *κῆδος* 'care, distress' etc.), note pl. *sādrā* (Y. 34.7, 43.11, 45.7);

5.3.6. [**h₂éi̯g-reh₂* ?; zero grade in Latv. *îgt* 'pine away'] Lat. *aegrum* [Plt.+] 'distress', usually taken as a substantivization of *aeger* 'sick', but (nominal) *aegro-* is clearly quite old – cf. Nussbaum's analysis of *aegrōtus* 'sick, pining' (1996:§2.1, 1998b:§1.2.1), a unique and possibly quite archaic formation in Latin;

5.3.7. [**pléh₂-reh₂* ?] OIr. *lár* 'surface, floor' (*'flat area'; root-shape **pleh₂-* in Li. *plóti* 'flatten'); cf. **pleh₂-ru-* in Gmc. (Eng. *floor* etc.);

5.3.8. [**kés-reh₂* ?] OIr. *cír* f. 'comb' (**kes-reh₂*), cf. Hitt. *kišri-* c. 'carding';

5.3.9. [**(h₁)rébʰ-reh₂* ?] RCS *rebro* 'rib', including pl. tant. *rebra* 'side' (e.g. at J. 19.34, translating *τὴν πλευράν*; Sreznevskij III.113 s.v.), perhaps from an original collective **(h₁)rébʰ-reh₂* 'covering, enclosure' (: Gk. *ἐρέφω* 'cover', *LIV* 448) > 'rib cage', with sg. *rebro* back-formed.[14]

6. Forms with a-vocalism.

6.1. If one operates, counter to "Leiden School" doctrine, with PIE **a*, forms like **R(a)-ro-** (cf. **nas-ro-*, §3.2. above) offer little surprise, since no (truly)

[13] According to Ruijgh (1995:347 with n. 7), Myc. (dat.) *te-qi-ri-jo-ne* presupposes the existence of an adj. **tʰekʷʰrós* (*τεφρός* as such otherwise not attested until the 4th. c.); but such an adjective could have been formed on the basis of a collective at any point; nor is there any reason to suppose (pace Ruijgh) that the collective continued by *τέφρα* replaces an original neut. **tʰékʷʰ-r̥* 'ash, cinder'.

[14] A suggestive parallel appears in *τὰ πλευρά* (also f. pl.) 'ribs, side', although this is generally taken (without much basis, as far as I can see) to continue a *u̯er(/n)*-stem.

old corresponding zero grades would exist anyway. Nevertheless, it would not be without interest if some such material could be understood as old collectives, as perhaps in cases like the following.

6.2. For **sak-ro-*, the adjective (as in Lat. *sacer*, U. **sakra** etc.) could be back-formed from an older **sák-reh$_2$* 'rites' (cf., in effect, Lat. *sacra facere*, and the old compound seen in *sacerdōs*, presupposing a **sakrom* or **sakrā̆* as a basis [Nussbaum 1999:396]); this, in turn, could be the source of the widespread early denominative **sakreh$_2$-i̯e/o-* (Lat. *sacrāre*, O. **sakarater** etc., unless this is a factitive of the *nouāre* type built on the adj. **sak-ro-*, or else a relatively late denominative built to **sakro-* with productive **-āi̯e/o-*, cf. *dōno-/dōnāre* etc.).

6.3. Lat. *faber* 'artisan', Arm. *darbin* 'smith', and (with somewhat less certainty) OCS *dobrŭ* 'good, beautiful' are generally taken to point to a **d^{h}ab^h-ro-* (cf. Go. *gadaban* 'be fitting').[15] As an agentive term, Lat. *faber* does not appear to provide promising evidence for an original collective; an adjective *faber* 'skillful' exists, but appears late [Ovid+], and seems to be a secondary creation. Nevertheless, the adv. *fabrē* 'skillfully' is old [Plt.+]; conceivably, then, a collective **d^{h}áb^h-reh$_2$* 'handicraft, things that fit together' could have yielded a back-formed adjective **d^{h}ab^h-ro-* 'craft-working', whence easily both *fabrē* and (agentive) *faber*, cf. also the more or less predictable (Nussbaum 1999) parallel abstract **d^{h}ab^h-ri-* 'craft-working', which may underlie Arm. *darbin* (< **d^{h}ab^h-ri-no-*).[16]

6.4. It is usually assumed (e.g. Melchert 1993:107, 1994:88[17]) that Hitt. *kallar* 'baleful, unfavorable' reflects a basic **-ro-* adjective **ĝ^{h}alH-ro-*, beside which OIr. *galar* n. 'sickness, sorrow' shows a substantivization. But F. Starke (1990:355ff.) has argued that the Hittite adjectival use may actually be secondary, arising from predicatival use of *kallar* as a neuter noun, and possibly also from a misunderstood asyndetic noun phrase *kallar idālu uttar*, whence *kallar* in attributive usage in the phrase *kallar uttar*.[18] If this is so, then Hitt. *kallar* and OIr. *galar*

[15] M. Huld (1995) adds the Messap. titular term *tabara*, based on his assumptions about Grassmann's Law in Messapic.

[16] CLuv. *tapar-* 'rule, govern' and perhaps also the Hittite kingly title *Tabarna/Labarna* point to a **d^{h}ab-ro-* or **d^{h}ab^h-ro-* "strong, mighty' that belongs semantically with the **d^{h}ab-ro-* of OHG *tapfar*, NHG *tapfer* 'brave, gallant', according to a widely-accepted etymology of H. Eichner (for references and discussion, see recently Melchert 1997:87f., Katz 2001:220 n. 42). It is not clear to me, however, that this material belongs with Lat. *faber* etc.

[17] Q.vv. for discussion of additional **-ro-* material in Anatolian.

[18] In a letter to me of 1 June 1999, Craig Melchert indicated his tentative acceptance of Starke's position (despite his earlier pronouncements *contra*, locc. cit.), and provided me with extremely helpful references and commentary.

might point to an old neut. **ĝ^{h}alH-ro-m* '(instance of) sickness, sorrow etc.', which in turn could have a source in an old collective **ĝ^{h}alH-reh$_2$* 'sickness, sorrow etc.'.

6.5. Beside the **stag-*[19] of στάζω 'drip; drop': OBret. *staer* f. 'river, brook' (Celt. **stagrā* < collect. **stág-reh$_2$* 'dripping(s)' ?).

7. Possible applications to other material.

7.1. The general conception under consideration may help explain certain otherwise difficult material, such as the following.

7.2. Lat. *labia/labea*, *labium* and *labrum* all have the meaning 'lip', pl. 'lips', without any evident basis for the morphological variation – somewhat surprising in a body part term. It seems to be the case, however, that in transferred usages, i.e. referring to the 'edge' or 'brim' (of a vessel, ditch, etc.), it is mainly *labrum/labra* [Cato+] that predominates. While the vocalism of *lab-* seems clearly to derive from a zero grade,[20] the suffix formation of (pl.) *labra* may point to a collective **léb-reh$_2$* 'lip-age, lip-like things', with the vocalism of the resulting **lebra* inevitably adjusted after *labia* (the more direct continuator, in one way or another, of the actual body part term).

7.3. Among various Greek forms in need of explanation, two items may be addressed briefly here, while δηρός calls for more extended discussion (**§8.**).

7.3.1. The anomalous accent of γαῦρος [Archil.+] 'exulting in' (: [ptcple.] γαίων 'id.', γάννμαι 'be delighted at' etc.) is difficult to explain, but might derive from a collective **géh$_2$u-reh$_2$* 'exulting, exultation' (amounting essentially to a verbal noun), which could then have been adjectivized secondarily. (Cf. also n. 33 below.)

7.3.2. The **R(e)-ro-** form Hom. νεκρός 'corpse' has the same (substantival) meaning as νέκυς 'id.' (cf. Av. *nasāu-*), whereas the adjectival sense 'dead' occurs once in Pindar (fr. 192) and is otherwise late. The antiquity of the meaning 'corpse', moreover, seems guaranteed by Myc. (KN) *ne-ki-ri-de* (f. nom. pl.), *ne-ki-ri-si* (dat. pl.), if this term refers (as is generally assumed) to 'shroud weavers' or the like. Thus one could imagine a collective **nék̑-reh$_2$* 'dead material', secondarily animatized (> 'the dead') and provided with a back-formed singular ('corpse'). Given the existence of a root noun **nek̑-* (as in Lat. *nex*, νέκες Hsch.), this formation could be a secondary one of the type discussed in §8. (cf. **§8.1.** μηρός). There is no reason to believe, with Ruijgh (1995:347), that νεκρός was modeled on the

[19] *LIV* 538 ***steh$_2$g-**, but admitting "möglich auch **stag*".

[20] Cf. Eng. *lip* and other Germanic material, which may point to an original root noun **leb-* or *s*-stem **léb-(e)s-* (Wagner 1997), either of which could provide a zero-grade allomorph; on *lab-* as zero grade, see the further references in my earlier discussion of such forms, 1999:13. (Differently Hamp 1998:337f., claiming IE **lo-* > Lat. *la-* regularly, but cf. Schrijver 1991:475f.)

**t^{h}ek^{wh}rós* that allegedly underlies Myc. *te-qi-ri-jo-ne* (cf. n. 13 above), as a substitution for the phonologically expected **akrós* < (morphologically regular zero-grade) **n̥kró-*. The related idea (1995:347n8) that this **akrós* would have been homophonous with **akrós* 'sharp' founders on the possibility that the latter may never have existed, if (cf. §4.4.) ἄκρος was built directly to ἄκρα, and not (as Ruijgh believes) as a replacement of an original zero-grade adjective (cf. n. 12 above). Even if the Caland-system associations of the root seem to validate the traditional assumption that νεκρός is nothing more than a substantivized **-ro-* adjective,[21] the full grade is surprising (cf. zero-grade **n̥k-* in OIr. *éc* 'death' and perhaps Toch. B *enkwe*/A *onk* 'man' < *'mortal'). Failing other attempts to explain this secondarily (as per Ruijgh above, or perhaps – though not, to my mind, satisfactorily – on the basis of the root noun otherwise poorly attested in Greek), it becomes at least possible to consider other factors, such as the full-grade collective entertained here, with all due reserve.

8. δηρός.

8.1. Especially if δηρός and Arm. *erkar* form an exact equation pointing to a potentially old **dṷeh$_2$-ro-*, as is usually assumed,[22] the reason for the creation of a full-grade formation (beside the inherited zero-grade **-ro-* adjective **duh$_2$-ró-*, §1.1.) is far from obvious.[23] Perhaps the best assumption[24] is that **dṷeh$_2$-ro-* (or at least δηρός) owes its full grade to the root noun seen in δήν (cf. §2.2.); but while such an observation accounts mechanically for the vocalism, it does not actually explain the derivation. Nor is anything really explained by the traditional assumption that **duh$_2$-ró-* essentially refers to "space" (Ved. *dūrá-* 'far') while **dṷeh$_2$-ro-* essentially refers to "time" (δηρός, Arm. *erkar* 'for a long time'), even if this distinction (which could easily have arisen secondarily) accurately reflects something old, a claim that is open to some doubt.[25] Most seriously, it has emerged rather conspicuously from the above survey that with certain limited and possibly secondary exceptions,[26] full-grade **R(E)-ro-** adjectives, insofar as they exist, do not tend

[21] See in these terms Weiss 1993:142n39.

[22] Predictable doubts in Clackson (1994:112ff.), based on Arm. *erkayn* 'long, wide' (of distance), which may suggest (but surely does not prove) that *-ar* in *erkar* is a relatively late suffixal addition.

[23] Full grade in and of itself is not problematic, since this is otherwise attested (e.g. OIr. *doe* 'slow' < **dṷoh$_2$i(j)o-*, cf. also Arm. *erkayn* < **dṷeh$_2$ino-*, if the formation is old); it is the apparently isofunctional **-ro-* forms **dṷeh$_2$-ro-* : **duh$_2$-ró-* that call for explanation.

[24] Thus J. H. W. Penney apud Clackson 1994:46.

[25] The distinction breaks down with forms like Lat. *dūrāre* ("time") and Arm. *erkar* (which sometimes refers to "space"; Meillet 1924:4).

[26] See e.g. §10.3.3. below on **bhid-ro-*/**bhoid-ro-* in Germanic.

to have ordinary **R(Ø)-ró-** counterparts. Instead, **R(E)-ro-** forms appear to be largely substantival (with a suggestive concentration of collective-looking material), and at least some **R(E)-ro-** adjectives are susceptible of being interpreted as back-formed to substantives.

8.2. For a "pure adjectival" **-ro-* form (whether primary, i.e. based on a root meaning 'be extended', or secondary to a root noun meaning 'extent'), one expects a straightforward adjectival meaning like 'extended'. While this is indeed what the Armenian form seems to show, δηρός is not really adjectival as such in early epic. Of 48 Homeric occurrences (see *LfgrE* s.v.), fully 44 show what is normally assumed to be adverbial δηρόν; the remaining four instances all involve an attributive usage in the phrase δηρὸν χρόνον (in one repeated Iliadic line and two occurrences in Homeric hymns), suggesting that attributive δηρός is a late development.

Among the various constructions identified in the *LfgrE* survey, the unique "predicative" use at I 415 is particularly interesting and potentially important.[27] In addition to uniquely displaying this syntactic feature, this is the only Iliadic instance of "respected" digamma for this form (digamma being otherwise neglected, with the sole exception of α 203, in a passage patently based on I 414-5). The passage in question occurs in the embassy to Achilles, embedded within a famous κλέος ἄφθιτον passage, which must surely be very old:

412 εἰ μέν κ' αὖθι μένων Τρώων πόλιν ἀμφιμάχωμαι,
413 ὤλετο μέν μοι νόστος, ἀτὰρ κλέος ἄφθιτον ἔσται·
414 εἰ δέ κεν οἴκαδ' ἵκωμι φίλην ἐς πατρίδα γαῖαν
415 ὤλετό μοι κλέος ἐσθλόν, ἐπὶ δηρὸν δὲ μοι αἰὼν
416 ἔσσεται, ...

More specifically, the predicative usage 'life(time) will be (left) for me' incorporates another unique feature, namely the prepositional construction ἐπὶ δηρὸν. In effect, then, the passage literally conveys an expression like 'life(time) for me will be for the duration' – and not surprisingly, Chantraine (1953:111) actually classifies this instance among examples of ἐπὶ in prepositional usage with *nouns* expressing duration (cf. B 299 μεῖναι' ἐπὶ χρόνον, θ 226 οὐδ' ἐπὶ γῆρας). If this isolated and clearly archaic expression[28] is taken at face value, it may point to an originally substantival **dṷéh$_2$-ro-m* 'extent, duration', mainly used adverbially, with

[27] Cf. §6.4. above on the predicative use of Hitt. *kallar*.

[28] Cf. also the v.l. ἐπὶ δηρὸν (for ἔτι δηρὸν) at P 41.

the nominal behavior at I 415[29] as a relic.[30] It is an open question as to how one might best conceive of the derivational process that underlies such a substantive: one need not, of course, assume a back-formed singular to a hypothetical collective **dy̯éh₂-reh₂*,[31] but one is at least freed from the necessity of accounting for parallel adjectives **dy̯eh₂-ro-* : **duh₂-ró-*, which seems not to be an attractive construct.

9. Concluding morphological remarks.

9.1. The conventional conception according to which "[o-stems] have no ablaut" (cf. §1.1.) requires qualification. In fact, a number of studies have sought to identify, for some stage of the protolanguage, something like root apophonic alternations (generally with concomitant accentual contrasts) in some plain thematic forms and in a range of suffixed thematic formations. What seems most striking about these patterns is that one of the members of these apophonic pairs is frequently a collective or feminine abstract formation, as in the following:

(i) **k^{w}é-k^{w}l(H)-o-* 'wheel' ~ collect. **k^{w}ek^{w}l(H)-éh₂* (Eichner 1985:139ff.) or (with a different accentual relationship) **k^{w}ek^{w}l(H)óm* ~ **k^{w}é-k^{w}l(H)eh₂* (Rasmussen 1999:§4.1);

(ii) **R(E)-mó-** (action nouns) ~ **R(Ø)-méh₂** (Hamp 1982, 1988) (here also adjectival **R(Ø)-mó-**);

(iii) **R(é)-tro-** ~ **R(Ø)-tréh₂** (Olsen 1988);

(iv) **R(é)-no-** ~ **R(o)-néh₂** (Barton 1993, **y̯és-no-/*y̯os-néh₂*; here also perhaps **y̯ḗs-no-*) (beside adjectival **R(Ø)-nó-**).[32]

9.2. Given the enormous range and quantity of IE **-ro-* material, the present study has relied, for heuristic purposes, on a relatively limited data base. Not only do we abjure anything like exhaustive coverage, but also freely acknowledge the tentative and somewhat programmatic nature of the presentation and the conclusions to be drawn from it. Nevertheless, the weakest claim that may be justified is that **R(E)-ro-** formations seem to show little connection with corresponding primary zero-grade **-ro-* adjectives, despite the superficial appearance of δηρός beside IE **duh₂-ró-*, or with original adjectival formations themselves. Instead, **R(E)-ro-** formations seem to be essentially substantival, including what appears to be non-negligible evidence for collective formations **R(é)-reh₂** (cf. ἄγρα, ἄκρα, τέφρα, ἕδρα, λέπρα; in part secondarily adjectivized, cf. ἄκρος, Λεπρός/λεπρός). The overall pattern is similar to some instances of secondary **-ro-* based on root nouns (μῆρα) and other secondary formations (**témH-s-reh₂*), and seems quite distinct from at least some other substantival primary **-ro-* formations, such as instrument nouns of the type Ved. *kṣurá-* 'razor' (: ξυρόν 'id.'). The relation of such **R(é)-reh₂** collectives to patterns like those in §9.1. deserves further study.

10. Some remaining problems.

10.1. In keeping with the nature of this treatment (§9.2. ad init.), it may be appropriate to conclude by signaling some of the problems affecting **R(E)-ro-** forms that remain to be elucidated in full (some already noted above, passim), at both the PIE (§10.2.) and einzelsprachlich (§10.3.) levels.

10.2.1. An immediate question concerns the factor of root structure. It could certainly be claimed (cf. Ruijgh on τέφρα, n. 13 above) that at least some **R(E)-ro-** full grades are merely artifacts of root-structure constraints against zero grades of double-obstruent CeC roots (cf. **pek^{w}-* 'cook': **pk^{w}-tó-* → /*pek^{w}-tó-/ et sim.). While this may account for some forms, it does not account for nearly all of them, and would not in any event apply to root-accented collectives. If the latter, moreover, do reflect an originally distinct morphological category, this might help resolve certain accentual difficulties displayed by some of the material (cf. §4.1. on Ved. *ájra-* : ἀγρός).[33]

10.2.2. While some alleged cases of overt *e/o*-ablaut (or at least parallel *e/o* derivations) in *ro*-stems are doubtful or at best inconclusive (e.g. ἔϋληρα/Lat. *lōra* 'reins', on which see Peters 1980:49ff.), there is at least suggestive evidence for such a pattern – however it is to be interpreted ultimately – in a few forms. A clear case seems to be **meh₁-* 'great' (comparative and superlative in Eng. *more, most*), which shows an *e*-grade **meh₁-ro-* (e.g. OHG names in *-mār*: *Volkmār* etc.,

[29] Perhaps also in the rare usage δηρὸν μένω (B 298 αἰσχρόν τοι δηρόν τε μένειν 'it is disgraceful to wait "the duration"')?

[30] In this event the oxytonesis of δηρός (vs. original **δῆρον*) would be secondary, as a more or less predictable artifact of the later attributive usage (already established, as seen above, in the epic period). The process is at least mildly productive within Greek, cf. e.g. χόνδρος [Aristoph.+] 'groats, gristle' → adj. χονδρός [Hipp., Aristotle+] 'granular, coarse'. (See further n. 33 below.)

[31] Aliter, e.g.: primary substantival **-ro-* of the **deh₃-ro-* type (cf. §2.6), substantival vr̥ddhiform **dy̯éh₂-ro-* 'that which is long' formed to **duh₂-ró-* 'long', etc.

[32] A more far-ranging theory of contrastive accent that unites some of this material (as well as other thematic and some athematic formations) is presented in Rasmussen 1999.

[33] After this paper was drafted for its original presentation, I was made aware (thanks to Joshua Katz) of the interesting work of P. Probert on the accentuation of Greek words in *-pos*; see the working paper published as Probert 1998 (in part superseded by more recent and as yet unpublished findings, as the author informed me in some helpful communication). Some of the primary concerns of Probert's work overlap in interesting ways with material discussed above (such as Probert's attention to unexpected recessively-accented forms), even if the details of the analysis inevitably differ in some cases, partly as a result of Probert's more synchronic focus.

also the denominative source of Go. *merjan* 'proclaim [something great]' etc.) beside an *o*-grade **moh$_1$-ro-* (e.g. in Celt.: OIr. *már* 'great' etc., Gaul. names in *-māros* [*Nertomāros* etc.]). (Cf. also §10.2.4. (iii) below.)

10.2.3. While **-ro-/*-ri-* derivational patterns are now somewhat clear (cf. §2.3.2., with references), the more rare **-ro-/*-ru-* pattern(s?) is less so; see e.g. Pinault 1997:228f. for recent discussion.

10.2.4. A full-scale study of the **R(E)-ro-** data might reveal more than one sort of semantic specialization among such forms (cf. already various semantic classifications of the Indic material proposed in Frisk 1934), the precise import of which (e.g. in morphological terms) would then need to be assessed. Thus, it may not be coincidental[34] that color terms seem to be relatively frequent, e.g.:

(i) **b^{h}leh$_3$-ro-*: Lat. *flōrus* 'yellow, blond' (Schrijver 1991:177), unless this is back-formed to *flōrēre* (Leumann 1977:554);

(ii) **ḱeh$_1$-ro-* (?): Skt. *śāra-* 'motley, dappled', *κηρύλος* 'halcyon (bird)' (*IEW* 582, but see *EWAia* III.489 s.v. *śāra-*);

(iii) **ḱei̯-ro-* (MIr. *cíar* 'dark brown') beside **ḱoi̯-ro-* (OE *hār* 'gray, *hoar(y)*', OCS *sěrū* 'gray'; *IEW* 541);

(iv) **mod^h(-)ro-* or **mad^h(-)ro-*: Slav. **modrŭ* (Cz. *modrý* 'blue' etc.), Gmc. **madra-* (OE *mædere* 'Rubia tinctoria, *madder* (used as dye)' etc.) (*IEW* 747, but see *ESSJ* s.v., 19:101-4);

(v) **neg^w-ro-*: one or more of *νεβρός* 'fawn', Arm. *nerk* 'color', Lat. *niger* 'black' (recent discussion in Janda 1996, cf. §1.2.);

(vi) **perḱ-ro-* 'dappled': Lat. *pulc(h)er* 'beautiful' (< **pElkro-*, with liquid dissimilation; *IEW* 821);[35]

(vii) **slEi̯(H)-ro-*: cf. the **sloi̯(H)-ri-* presupposed by Lat. *lūridus* 'bruise-colored, pale yellow' (Nussbaum 1999:403f.).

10.3.1. Among problems that have a more einzelsprachlich cast, the restriction affecting IIr. full grades of the type *candrá-* etc. has already been mentioned (§1.2. above), and is likely to involve one or more types of secondary generalizations within the IIr. period, though possibly based on something old. For example, *candrá-* 'brilliant' itself appears beside Lat. *candēre* 'burn, glow', in which case *candrá-* would appear to be a Caland-type **-ro-* adjective, for which original zero grade is expected. But if Schrijver's conception of the *a*-vocalism of forms like *candēre* is correct (1991:425ff., 428 on *candēre*; cf. Meiser 1998:82f.),

[34] Even in connection with the idea of original abstracts/collectives: cf., for example, Myc. *e-ru-ta-ra-pi* (KN, 4x in the Ld series) 'with red (coloring)', pointing to a f. abstract /erut^{h}rā/ 'red(ness)' not otherwise attested in Greek; Hajnal 1995:148f.

[35] But this etymology of *pulc(h)er* is less than certain, given (among other problems) early and specialized meanings 'strong' and (esp. of animals) 'full-bodied; ritually pure' (see *DELL* s.v.).

this may reflect *e*-grade **(s)kend-* directly, which is itself anomalous for a stative in **-eh$_1$-*; in that event, both **(s)kend-eh$_1$-* and **(s)kend-ró-* may be archaisms, however the unexpected full grade is to be explained.[36] If so, an inherited *candrá-* (: *candati*) might then have exerted influence on some other such forms (cf. RV *-tandra-* [: *tandate*], etc.).

10.3.2. Some **R(E)-ro-** items, especially in Indo-Iranian, pose phonological difficulties that seem to require special explanations, as with various "Lubotsky's Law" forms of the type Ved. *pajrá-* 'firm' (: **peh$_2$ĝ-*, *πήγνυμι* 'fasten' etc.), *vájra-* 'cudgel' (if with **u̯(e)h$_2$ĝ-*, *ἄγνυμι* 'break' etc.; Lubotsky 1981, see recently de Lamberterie 1996, with a somewhat different conception).

10.3.3. The appearance of *o*-grades in a number of Germanic (sometimes also Baltic) **R(E)-ro-** forms (cf. **smoi̯r(i)i̯o-*, §4.2.2.), either in isolation or beside *e*-grade or zero grade, has been noted at least since Osthoff 1881:163 (on Go. *baitrs* beside a more widely-attested zero grade, OE *bittor* '*bitter*' etc.). Although the *o*-grade could be old in a few such cases,[37] it seems likely that most such forms ultimately reflect contamination from parallel thematic stems without *r*-formant, although the details of the spread of this process remain to be worked out in full; see, however, Wagner 1998:170 on the pattern Gmc. **haida-/*haidra-* '*heiter*', **swanka-/*swankra-* '*schwank*', etc.[38]

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[36] E.g.: the **(s)kénd-e/o-* 'inflame' of Lat. *incendō* may be a "faktitives Oppositionsaktiv zu urspr. medialem 'leuchten, glänzen'" (*LIV* 502, 2. **(s)kend* n. 3); perhaps the original middle was an acrostatic **(s)kénd-tor.*

[37] Cf. Peters (1999:451n26) on *βόθρος* 'hole, trench': "alter *-o-* Vokalismus ist in *-ro-*Stämmen nicht unerhört", with reference to Peters 1980:90ff.

[38] In principle (cf. §§2.3.1., 10.2.3.), one could also envisage influence from parallel **R(o)-ri-** forms of the type **h$_2$óḱri-* (on which see Nussbaum 1999).

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